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Department Of Electronics & Communication Engineering

RTL to GDSII Design and Optimization of a Reversible Compressor Circuit

Batch - 10A
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Project Abstract: Optimization of a Reversible Compressor Circuit

This project bridges classical VLSI design reversible hardware, focusing on the efficient physical implementation of a reversible 4:2 compressor.

- The project will focus on designing a **Reversible 4:2 Compressor** using **reversible logic** to achieve efficient and low-power data compression.
- The circuit will be modeled in **Verilog (RTL)** and implemented through the complete **ASIC design flow**, progressing up to the **GDSII layout** stage.
- **Ancilla bits** (helper bits) will be optimized to reduce circuit area, power consumption, and overall design complexity.
- The design flow incorporates Vivado for RTL design and simulation, followed by synthesis and physical implementation using **Cadence tools (Genus and Innovus)** for Place-and-Route (PnR).

Introduction

Quantum-Inspired Logic

Leveraging reversible computation principles from quantum systems to drive energy efficiency in classical hardware design.

4:2 Reversible Compressor

Implementing an efficient data compression circuit using reversible logic gates where inputs equal outputs.

Ancilla Optimization

Helper bits maintain reversibility while being optimized to reduce circuit area, power consumption, and complexity.

RTL to GDSII Flow

Complete ASIC design journey from Verilog modeling through synthesis, physical design, to final layout fabrication.

Why Energy-Efficient Arithmetic Matters

Power Dominance

Arithmetic circuits consume disproportionate power in modern VLSI systems, limiting battery life and thermal budgets

Compressor Role

Compressors enable high-speed multiplication through efficient partial product reduction in DSP architectures

Circuit-Level Optimization

Architecture choices at gate level directly impact system-wide power efficiency and performance metrics

Problem Statement

- **Information Loss**

  Conventional digital circuits lose information during computation, leading to increased power consumption and heat dissipation.

- **Reversible Logic Solution**

  Reversible logic avoids information loss by maintaining bijective mapping between inputs and outputs, reducing energy dissipation.

- **Ancilla Challenge**

  Designing efficient reversible compressors requires minimizing ancilla bits to reduce hardware overhead and improve performance.

Existing Method

Conventional Approach

- Uses irreversible logic gates (AND, OR, XOR)
- High power dissipation due to information loss
- Larger chip area and increased heat generation
- No quantum-compatibility or energy recovery

1. "Irreversibility and Heat Generation in the Computing Process"

Author: Rolf Landauer

Publisher: IBM Journal of Research and Development

Link: https://ieeexplore.ieee.org/document/5392738 ↗

- ✓ Proves that **irreversible gates like AND/OR/XOR lose information**, causing **power dissipation**.
- ✓ Shows why classical logic generates heat (Landauer's Principle).

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2. "Logical Reversibility of Computation"

Author: Charles H. Bennett

Publisher: IBM Journal

Link: https://ieeexplore.ieee.org/document/1055349 ↗

- ✓ Explains how **irreversible circuits waste energy**, while reversible ones can recover it.
- ✓ Establishes the need for **energy-efficient, quantum-compatible circuits**.

Proposed Method

01

Reversible Logic Gates

Utilize quantum-inspired gates: Feynman, Toffoli, and Peres gates for bijective computation.

02

4:2 Compressor Design

Implement reversible compressor architecture with optimized ancilla bit allocation.

03

Ancilla Optimization

Minimize helper bits to reduce circuit complexity, area, and power consumption.

04

Performance Benefits

Achieve reduced power dissipation, smaller area footprint, and lower thermal output.

Block Diagram

- **Five Inputs**

A, B, C, D, Cin fed into compressor

- **Carry Logic**

Carry redistribution combines intermediate carries

- **Three Outputs**

Sum, Carry, Cout generated

- **Full Adders**

Reversible full adders used

VLSI Design Flow using CADENCE

1

RTL Design

High-level behavioral description using Verilog for the reversible compressor circuit.

2

Simulation & Verification

Functional correctness validated through comprehensive test benches and waveform analysis.

3

Synthesis

RTL translated to gate-level netlist, optimized for Power, Performance, and Area (PPA).

4

Physical Design

Floorplanning, placement, routing, and clock tree synthesis for optimal layout.

5

SignOff & GDSII

Final DRC/LVS verification and GDSII export ready for fabrication.

Software Used

1

DSCH

Reversible gate design and schematic capture

2

Vivado

Simulation and functional verification

3

Cadence Genus

Logic synthesis and optimization

4

Cadence Innovus

Place and route

Working Principle

Compression Operation

4:2 compressor reduces 5 inputs into 3 outputs

Arithmetic Equation

A + B + C + D + Cin = Sum + 2Carry + 4Cout

Implementation

- Two full adders used
- Carry redistribution combines intermediate carries
- Correct arithmetic operation ensured

```
xcelium> run
=====================================================
 A B C D Cin | Sum Carry Cout || Expected Actual
=====================================================
 0 0 0 0  0  |  0    0     0  ||    0       0
 0 0 0 0  1  |  1    0     0  ||    1       1
 0 0 0 1  0  |  1    0     0  ||    1       1
 0 0 0 1  1  |  0    1     0  ||    2       2
 0 0 1 0  0  |  1    0     0  ||    1       1
 0 0 1 0  1  |  0    1     0  ||    2       2
 0 0 1 1  0  |  0    1     0  ||    2       2
 0 0 1 1  1  |  1    1     0  ||    3       3
 0 1 0 0  0  |  1    0     0  ||    1       1
 0 1 0 0  1  |  0    1     0  ||    2       2
 0 1 0 1  0  |  0    1     0  ||    2       2
 0 1 0 1  1  |  1    1     0  ||    3       3
 0 1 1 0  0  |  0    1     0  ||    2       2
 0 1 1 0  1  |  1    1     0  ||    3       3
 0 1 1 1  0  |  1    1     0  ||    3       3
 0 1 1 1  1  |  0    0     1  ||    4       4
 1 0 0 0  0  |  1    0     0  ||    1       1
 1 0 0 0  1  |  0    1     0  ||    2       2
 1 0 0 1  0  |  0    1     0  ||    2       2
 1 0 0 1  1  |  1    1     0  ||    3       3
 1 0 1 0  0  |  0    1     0  ||    2       2
 1 0 1 0  1  |  1    1     0  ||    3       3
 1 0 1 1  0  |  1    1     0  ||    3       3
 1 0 1 1  1  |  0    0     1  ||    4       4
 1 1 0 0  0  |  0    1     0  ||    2       2
 1 1 0 0  1  |  1    1     0  ||    3       3
 1 1 0 1  0  |  1    1     0  ||    3       3
 1 1 0 1  1  |  0    0     1  ||    4       4
 1 1 1 0  0  |  1    1     0  ||    3       3
 1 1 1 0  1  |  0    0     1  ||    4       4
 1 1 1 1  0  |  0    0     1  ||    4       4
 1 1 1 1  1  |  1    0     1  ||    5       5
=====================================================
 TEST COMPLETED
=====================================================
Simulation stopped via $stop(1) at time 320 NS + 0
./tb_reversible_4_2_compressor.v:71         $stop;
```

Implementation Methodology

01

Design Reversible Gates

Develop Feynman and Toffoli gates as fundamental building blocks

02

Construct Full Adder

Build reversible full adder using gate primitives

03

Implement 4:2 Compressor

Assemble compressor from two full adders

04

Simulate Design

Validate functionality using comprehensive testbench

05

Synthesize Circuit

Generate netlist using EDA tools

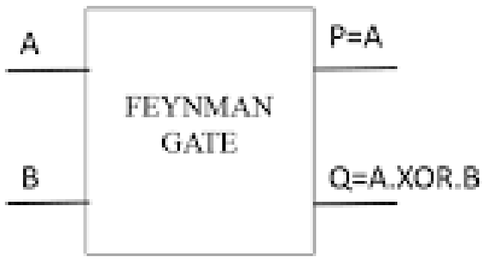
Reversible Gate Primitives

Feynman Gate

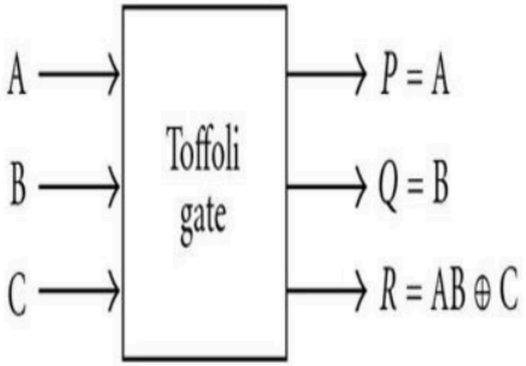
- Two-input, two-output gate
- Performs XOR operation
- Preserves input information
- Used for sum computation

Toffoli Gate

- Three-input, three-output gate
- Performs AND-based operation
- Generates carry signals
- Essential for arithmetic

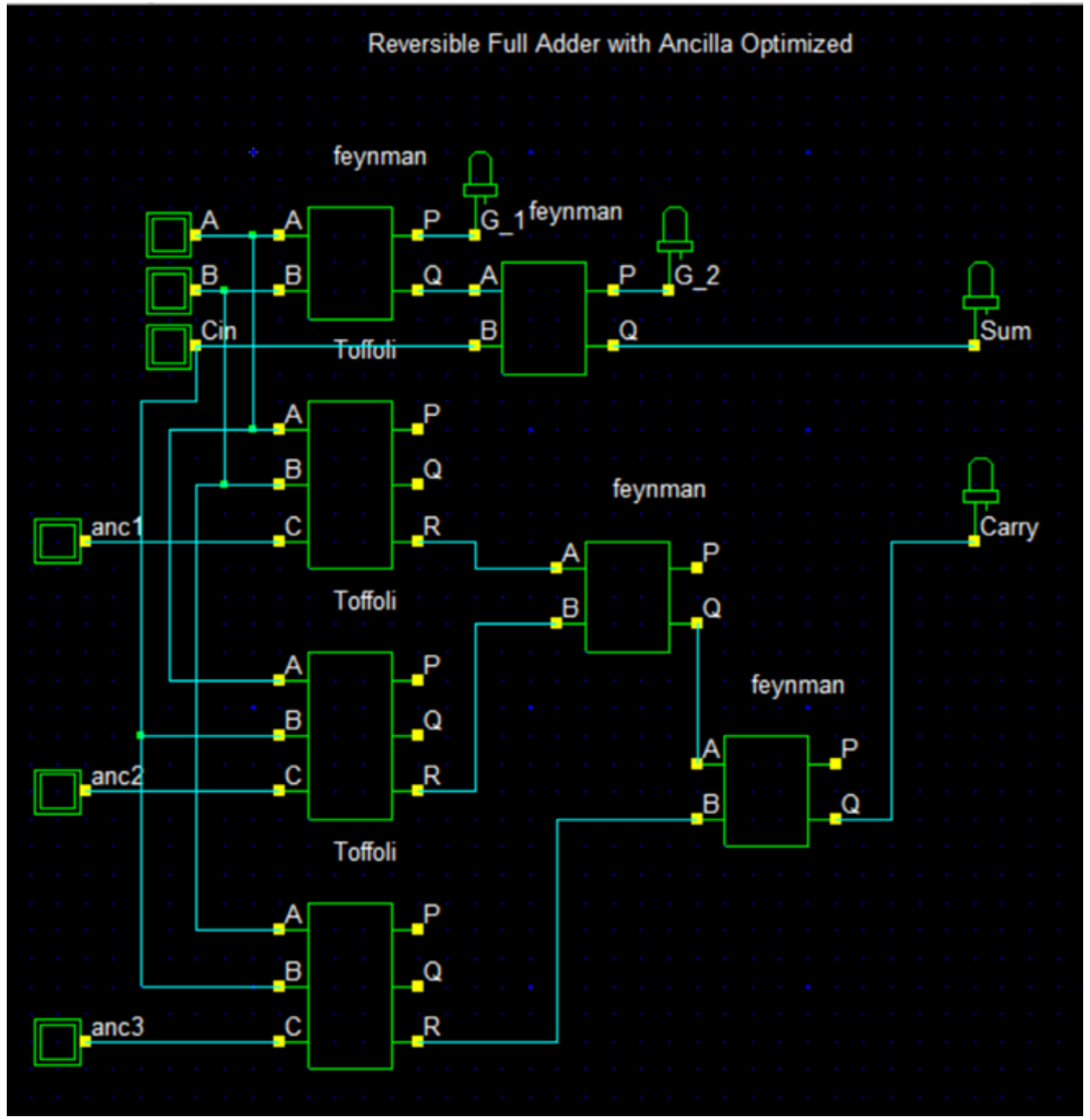


A	B	P	Q
0	0	0	0
0	1	0	1
1	0	1	0
1	1	1	1



Input			Output		
A	B	C	A	B	$(A \cdot B) \oplus C$
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

Reversible Full Adder Architecture



Key Characteristics

- Uses Feynman and Toffoli gates
- Computes Sum = A ⊕ B ⊕ Cin
- Carry generated via Toffoli gate
- Employs 1 ancilla input (constant 0)
- Produces garbage output for reversibility

Reversible 4:2 Compressor

Structure

Built using two reversible full adders connected in cascade configuration

Inputs

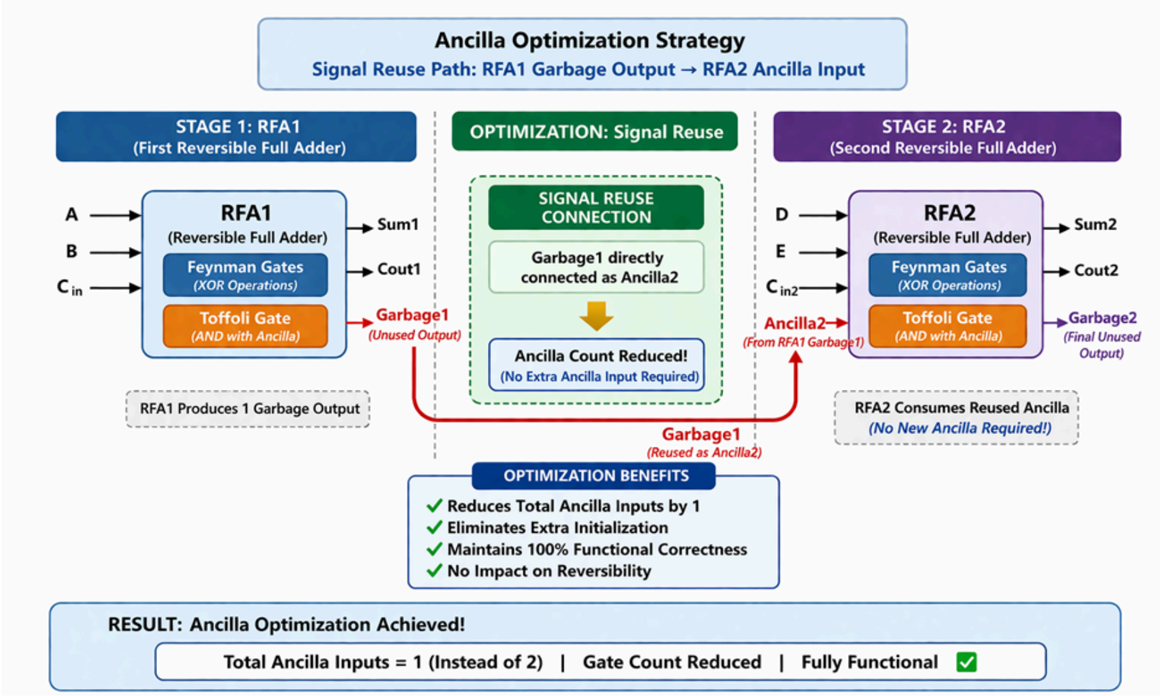
Four primary inputs (A, B, C, D) plus Cin for carry-in propagation

Processing

Intermediate carries (C1, C2) generated in two stages, then redistributed

Outputs

Final Sum, Carry, and Cout signals with garbage outputs



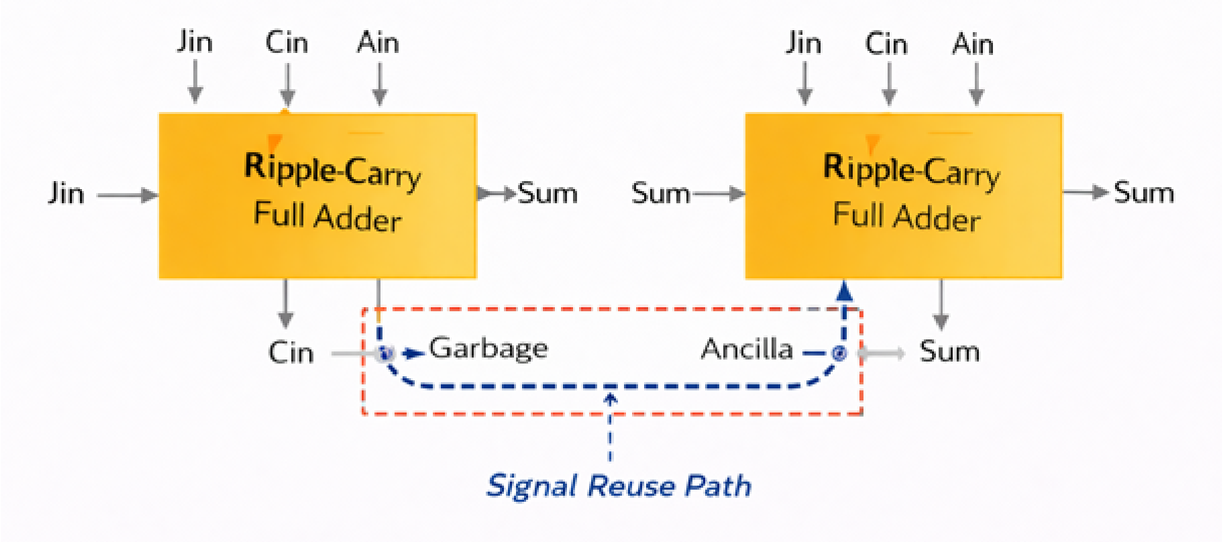
Ancilla Bit Optimization

Optimization Strategy

Minimize ancilla inputs through strategic signal reuse and efficient intermediate value management

Benefits

- Reduces circuit complexity
- Decreases area overhead
- Improves power efficiency
- Enhances overall design quality



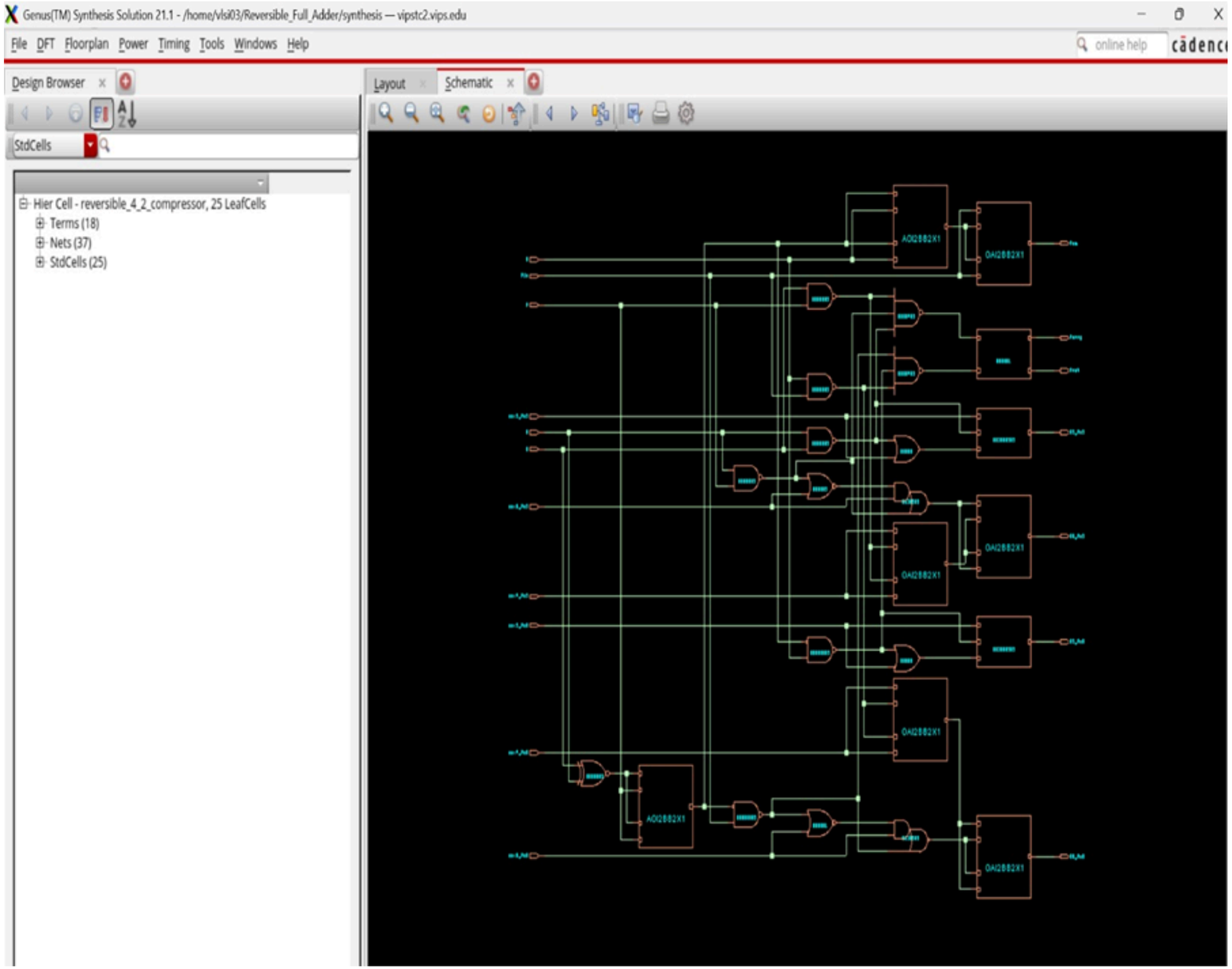
Simulation Results

Verification Summary

- 32 input combinations tested
- Expected vs. actual output match
- Functional correctness confirmed
- All test cases passed

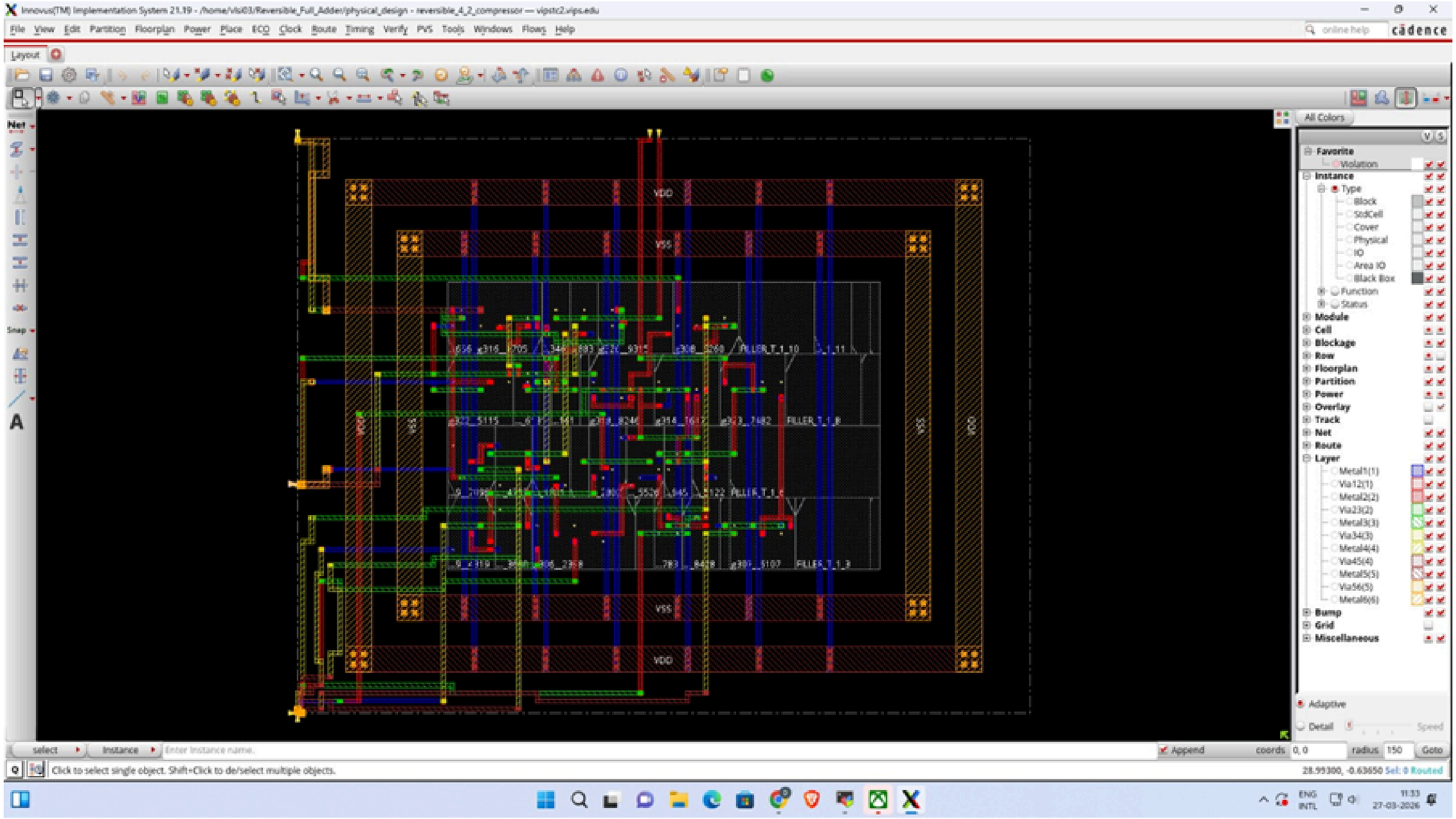
Synthesis Results

Cadence Genus Synthesis Report – Reversible 4:2 Compressor Standard Cell



PNR Results

Cadence Innovus Cell Placement View – Reversible 4:2 Compressor Floorplan and Placed Cells



Conclusion & Future Work

Key Achievements

- Successfully designed reversible 4:2 compressor
- Verified through simulation and synthesis
- Demonstrated zero information loss
- Established foundation for larger compressors

Advantages

- No information loss property
- Theoretical energy efficiency
- Quantum computing compatibility

Applications

- Quantum computing circuits
- DSP and multiplier architectures
- Low-power VLSI systems

Future Scope

- Extend to 8:3 compressor design
- Implement Peres gate optimization
- Complete physical place and route
- Measure power consumption

Current Research Status

1

Phase 1: Complete

- Designed and implemented conventional 4:2 compressor
- Verified functionality using Vivado simulation
- Generated baseline results for comparison (LUT, delay, correctness)
- Established reference model for reversible design evaluation

2

Phase 2: In Progress

- Designed reversible gates (Feynman and Toffoli)
- Constructed reversible full adder and 4:2 compressor
- Verified all 32 input combinations using testbench
- Performed synthesis using Vivado and Cadence Genus
- Validated zero information loss using output + garbage signals
- Compared reversible vs conventional design metrics

3

Phase 3: Complete (ASIC Implementation)

- Optimized area and logic efficiency of reversible design
- Successfully performed Place and Route using Cadence Innovus
- Completed full RTL-to-GDSII flow
- Generated final layout and verified design constraints
- Achieved timing closure and design rule compliance (DRC/LVS ready)

Thank You!!